CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 3: **The Atomic Foundations of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The formula unit mass of the compound $Ca_3(PO_4)_2$ is (Ca = 40, P = 31, O = 16): 1

(a) 262 u **(b)** 310 u **(c)** 230 u **(d)** 278 u

2. The empirical formula of a compound with C 85.7 per cent and H 14.3 per cent is: 1

(a) C_2H **(b)** CH_2 **(c)** CH_3 **(d)** CH

3. Assertion (A): A molecule of nitrogen has a triple bond. Reason (R): Each nitrogen atom needs three more electrons to complete its octet. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. The mass of 3 formula units of $NaCl$ relative to 1 formula unit of MgO (Mg = 24) is: 1

(a) 175.5 : 40 **(b)** 58.5 : 40 **(c)** 58.5 : 24 **(d)** 117 : 40

5. When 20 g of sulphur burns in oxygen producing 40 g of sulphur dioxide, the mass of oxygen used is: 1

(a) 40 g **(b)** 60 g **(c)** 10 g **(d)** 20 g

6. Assertion (A): Atoms are the smallest particles of matter which cannot be divided any further. Reason (R): Atoms consist of electrons, protons and neutrons. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. What mass of oxygen is present in 100 g of $CaCO_3$ (Ca = 40, C = 12, O = 16)? 1

(a) 16 g **(b)** 12 g **(c)** 32 g **(d)** 48 g

8. When sulphur and oxygen react in the ratio 1 : 1 to form SO_2 and 1 : 1.5 to form SO_3, these ratios illustrate: 1

(a) the law of multiple proportions **(b)** the law of conservation of mass
(c) the law of constant proportions **(d)** none of these

9. Assertion (A): Ionic compounds are hard and brittle. Reason (R): The ions in the lattice are held together by strong electrostatic forces and shifting a layer brings like ions together. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. How much calcium is present in 25 g of $CaCl_2$ (Ca = 40, Cl = 35.5)? 1

(a) 4.5 g **(b)** 11 g **(c)** 40 g **(d)** 9.0 g

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does an ionic compound dissolve in water but not in kerosene? 2

12. Calculate the molecular mass of $C_{12}H_{22}O_{11}$ (C = 12, H = 1, O = 16). 2

13. Calculate the mass of Na in 100 g of Na_2SO_4 (Na = 23, S = 32, O = 16). 2

14. State why covalent compounds like sugar do not conduct electricity even when dissolved in water. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the formation of the ionic compound Li_2O with electron transfer and the formula unit mass (Li = 7, O = 16). 3

16. State the postulates of Dalton's theory that explain 3
(a) the law of conservation of mass
(b) the law of constant proportions.

17. Explain why the law of conservation of mass is not violated in the burning of a candle even though it seems to lose mass. 3

OR

Calculate the number of atoms of each kind in 5 formula units of $Al_2(SO_4)_3$ and the total. 3

18. Calculate the masses of calcium and chlorine present in 55.5 g of $CaCl_2$. 3

OR

Compare the charge on the nucleus, the size and the electronic arrangement of Na and Na+. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Compare NaCl, MgO, CH4, H2O, diamond and iodine on type of bonding, nature of solid, melting point and conductivity in a table. 5

OR

Explain how the properties of substances such as table salt, sugar, graphite, diamond and copper depend on the nature of the bonding in them. 5

20. A mixture of 20 g of hydrogen and 20 g of oxygen is ignited. Find the mass of water formed, the mass of the excess reactant and the total mass after the reaction. Explain the use of the law of constant proportions. 5

OR

A mixture of two oxides of carbon is analysed. Sample A contains 12 g of carbon and 16 g of oxygen; sample B contains 12 g of carbon and 32 g of oxygen. Name the compounds, explain the law illustrated and predict the mass of oxygen with 12 g of carbon in a third, hypothetical oxide following the same pattern with ratio 1 : 3. 5

END OF QUESTION PAPER Total: 40 Marks